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ENCE 4620

5/4/26

Project 3—Morphological Image Processing

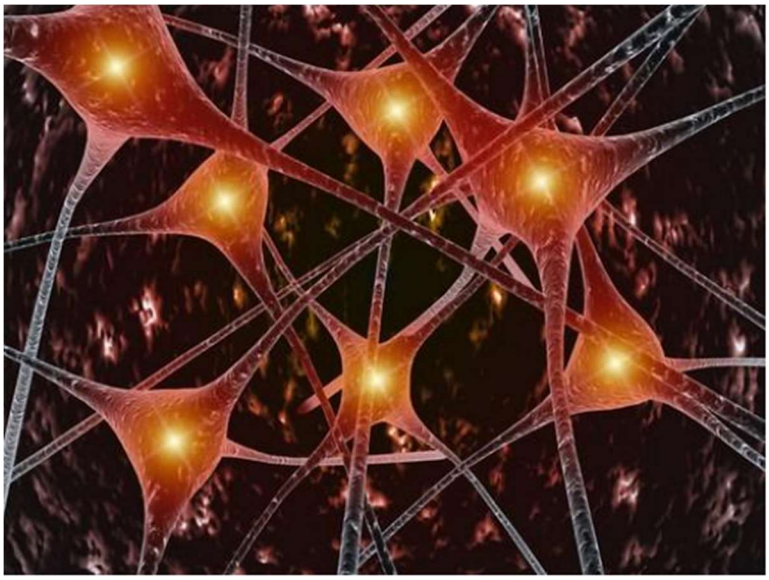
Introduction

The purpose of this project was to count objects and find centers of objects using binary masking and morphological image processing.

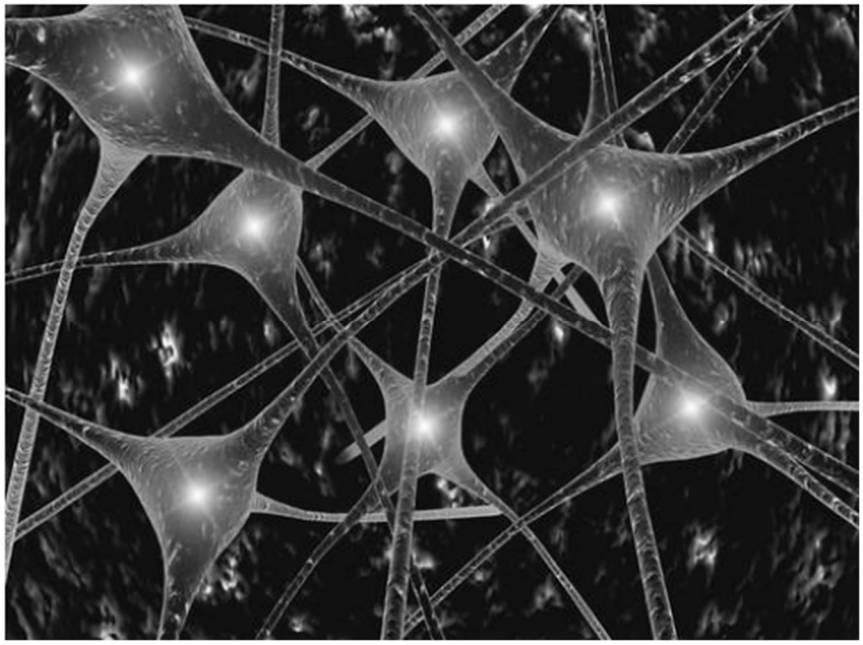
Method

1. Load the images
2. Convert to grayscale
3. Use Otsu's method to threshold the images
4. Process the images with morphological operations
5. Connected component analysis
    a. Finds all of the areas and centroids
6. Filter the components out by area
7. Complete the final count

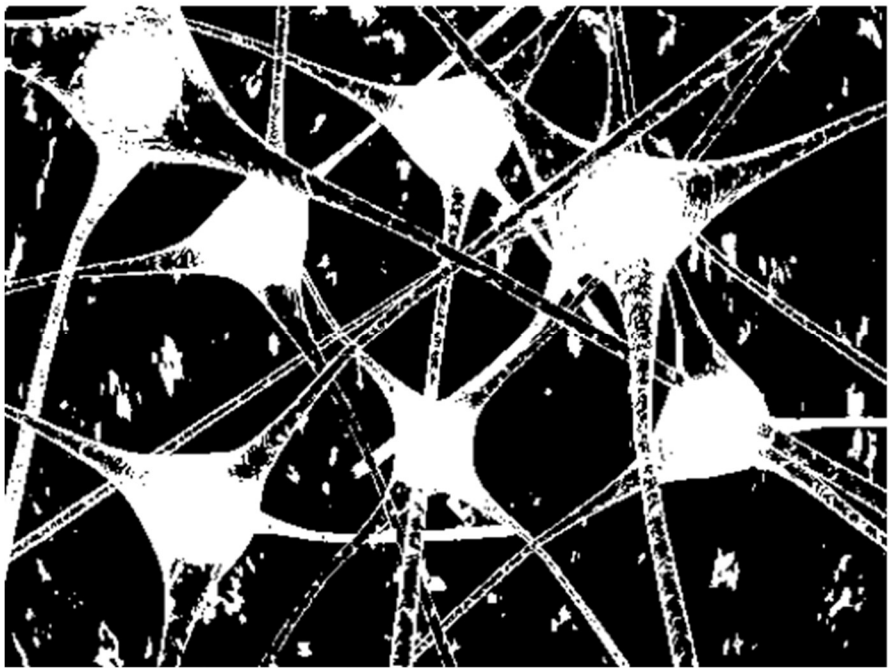
Image 1



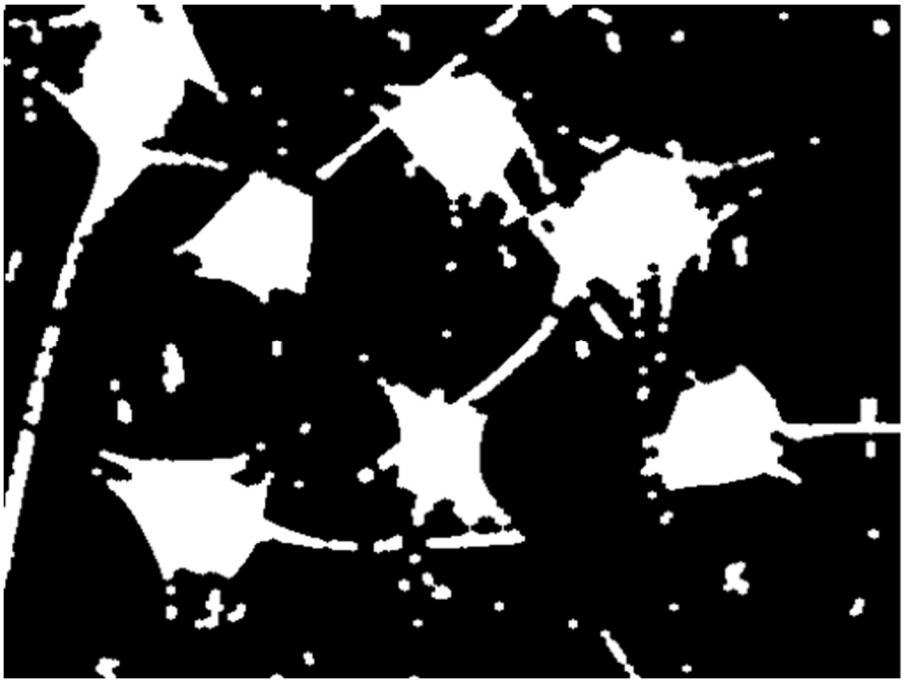
Grayscale:



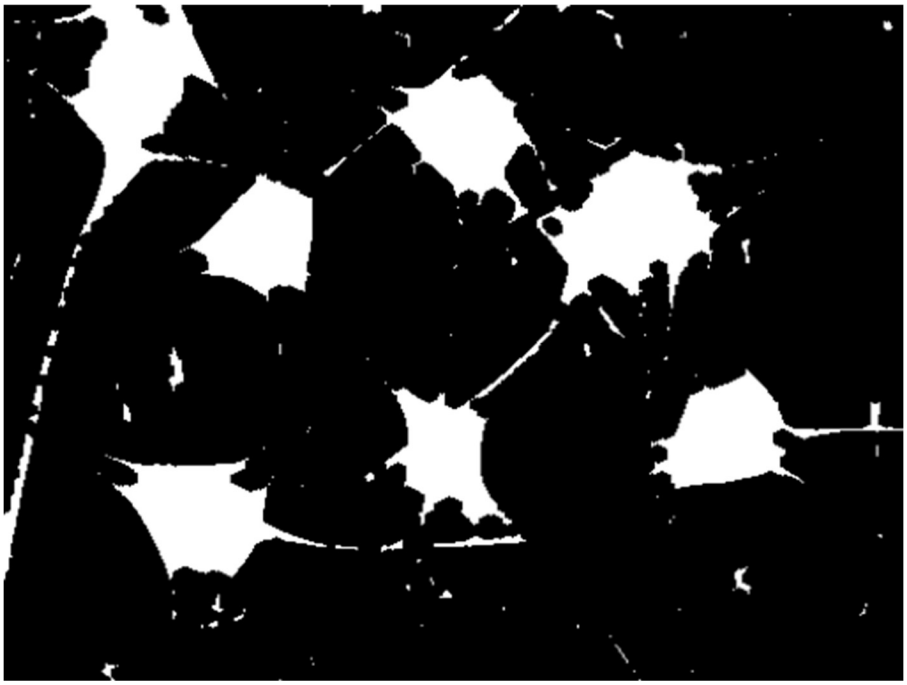
Binary mask:



After opening and closing:



After eroding:



The final count:

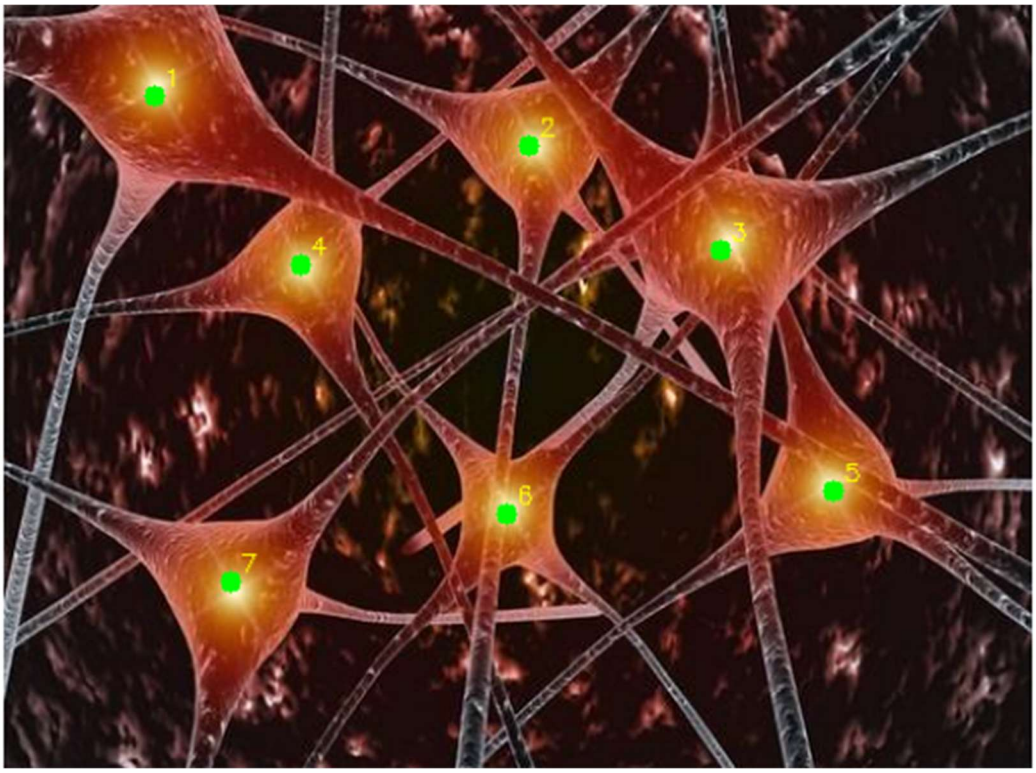
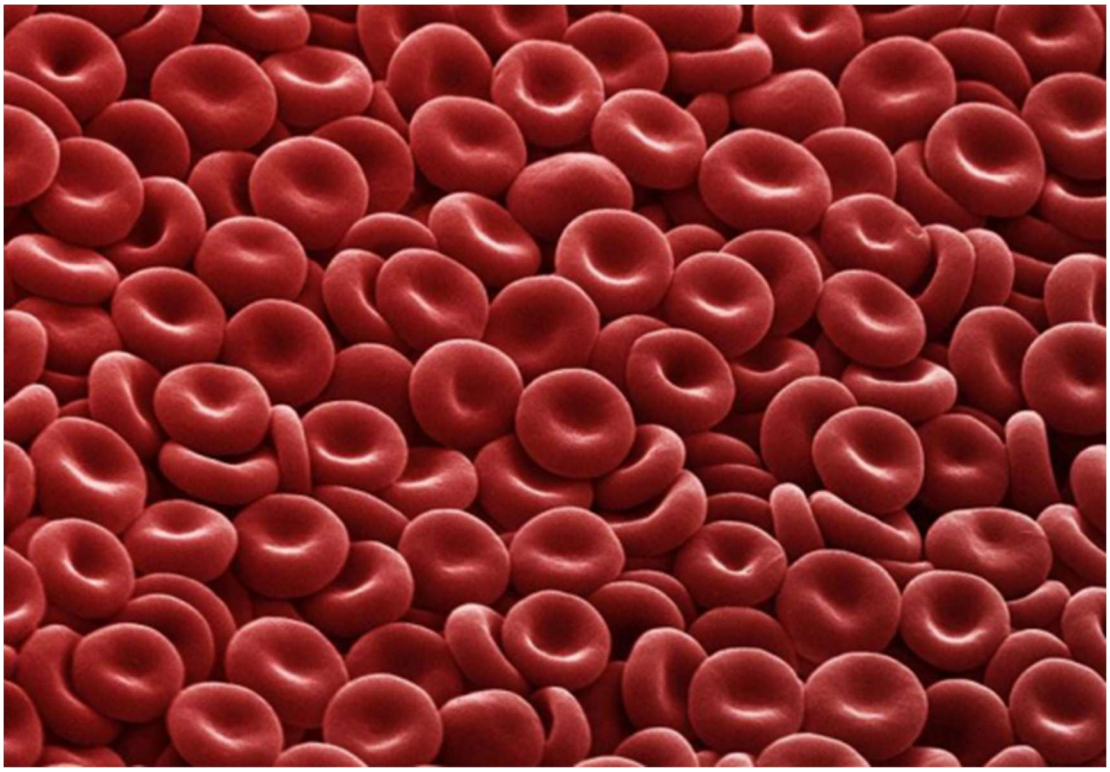
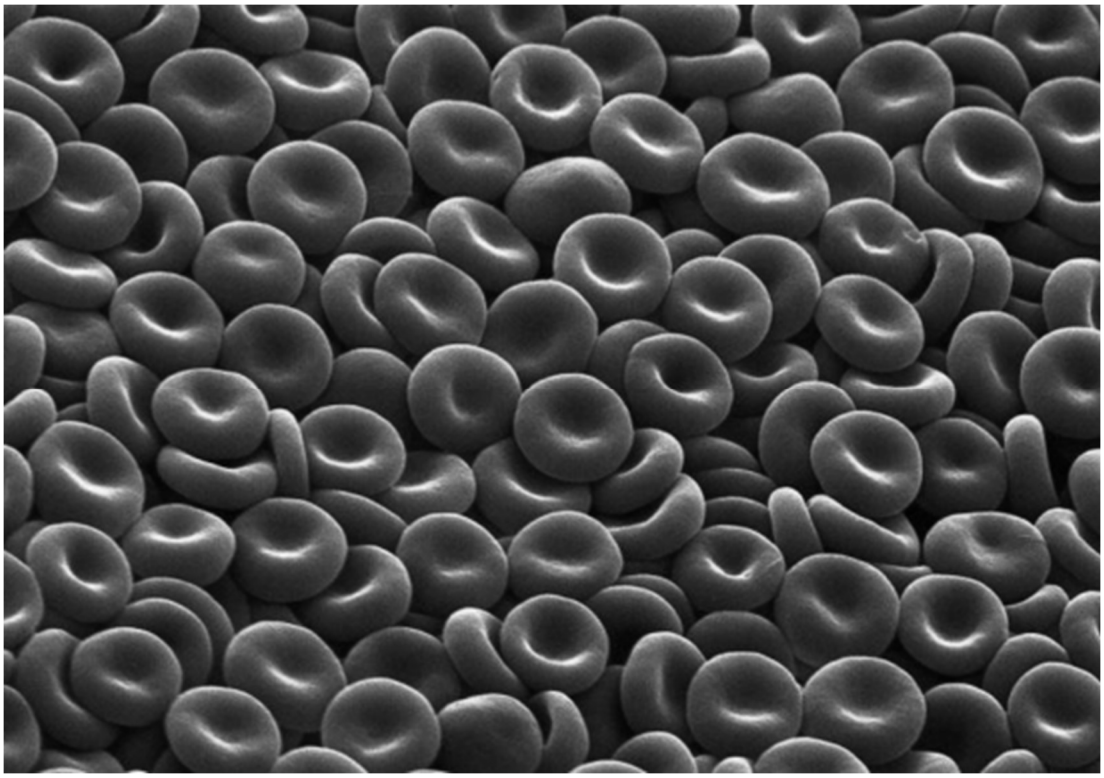


Image 2

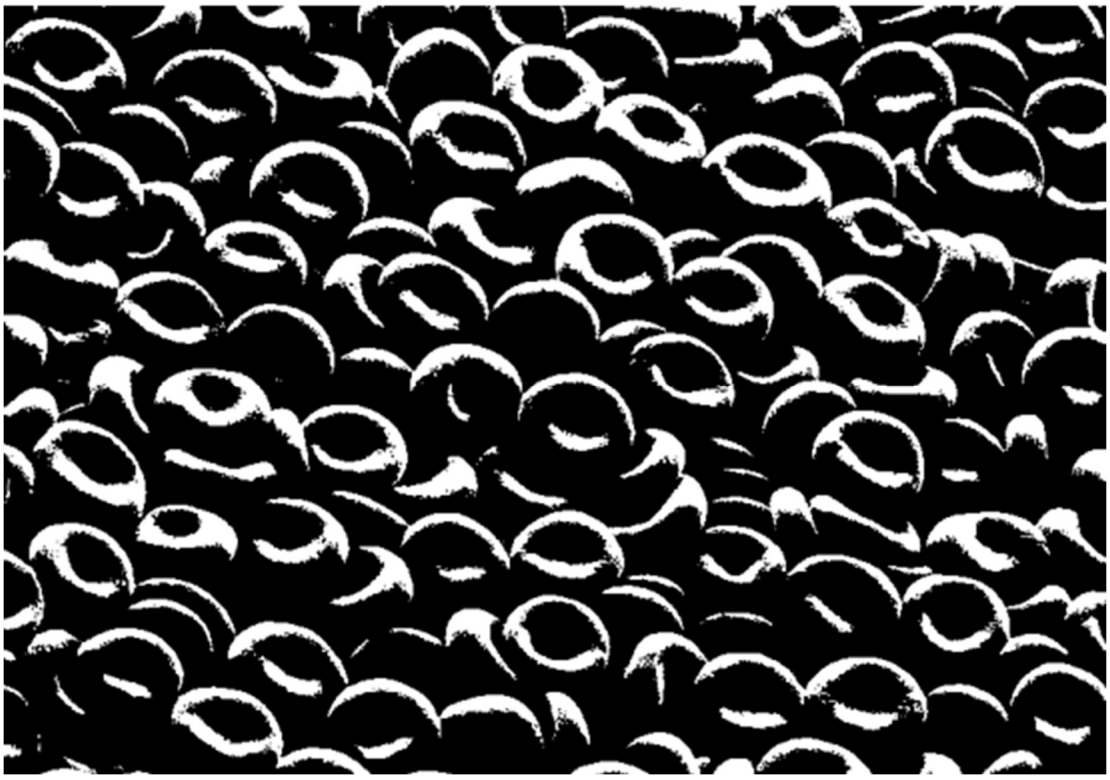
The structuring elements and the min/max areas were changed for this one, but I wasn't able to eliminate all double counting and missed cells. I'm honestly not sure what else I can do with this image, I tried inverting the mask as well.



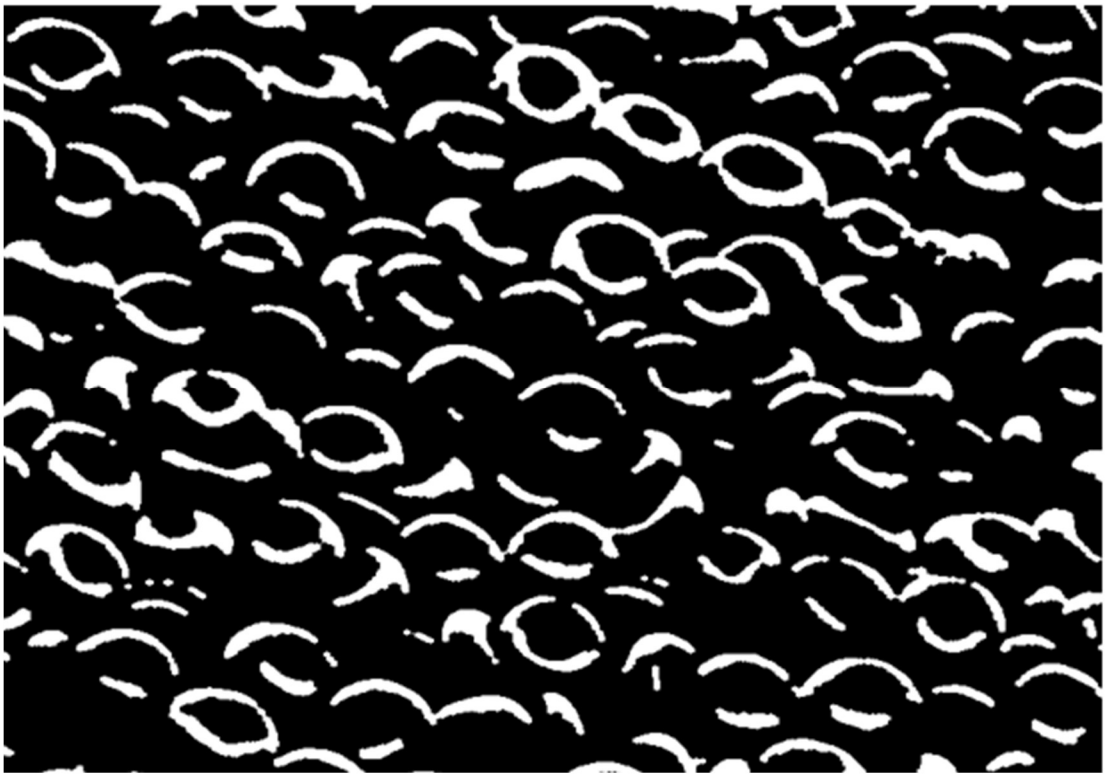
Grayscale:



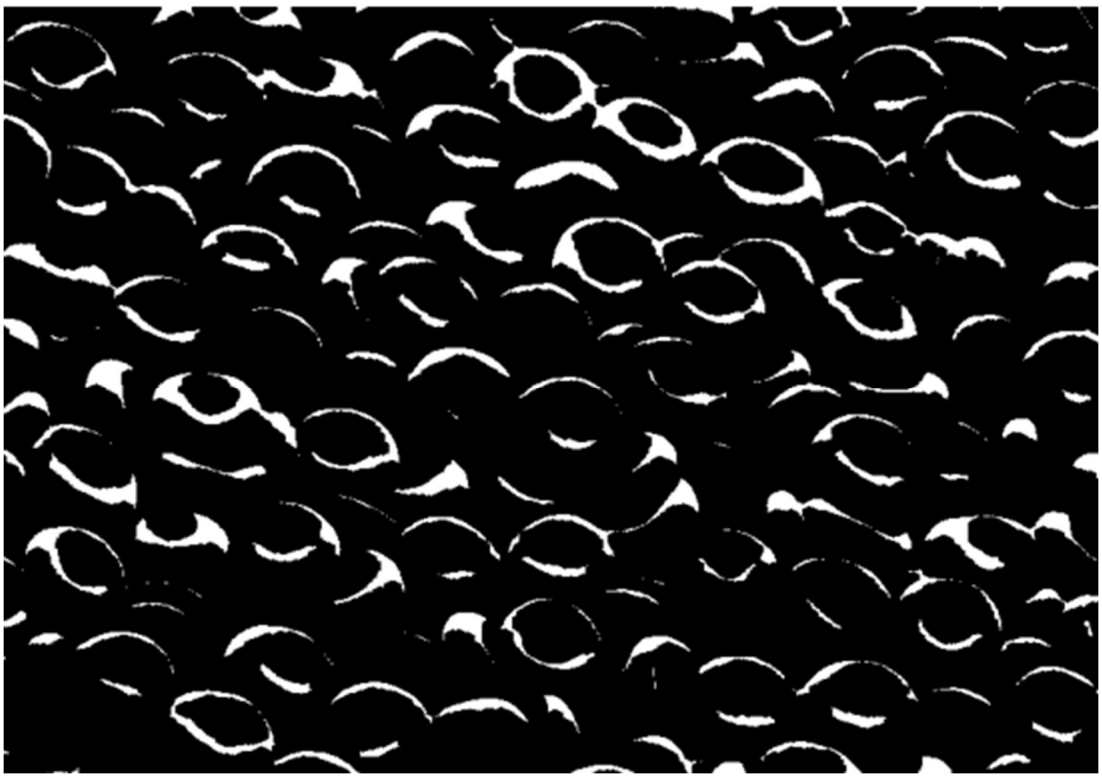
Binary mask:



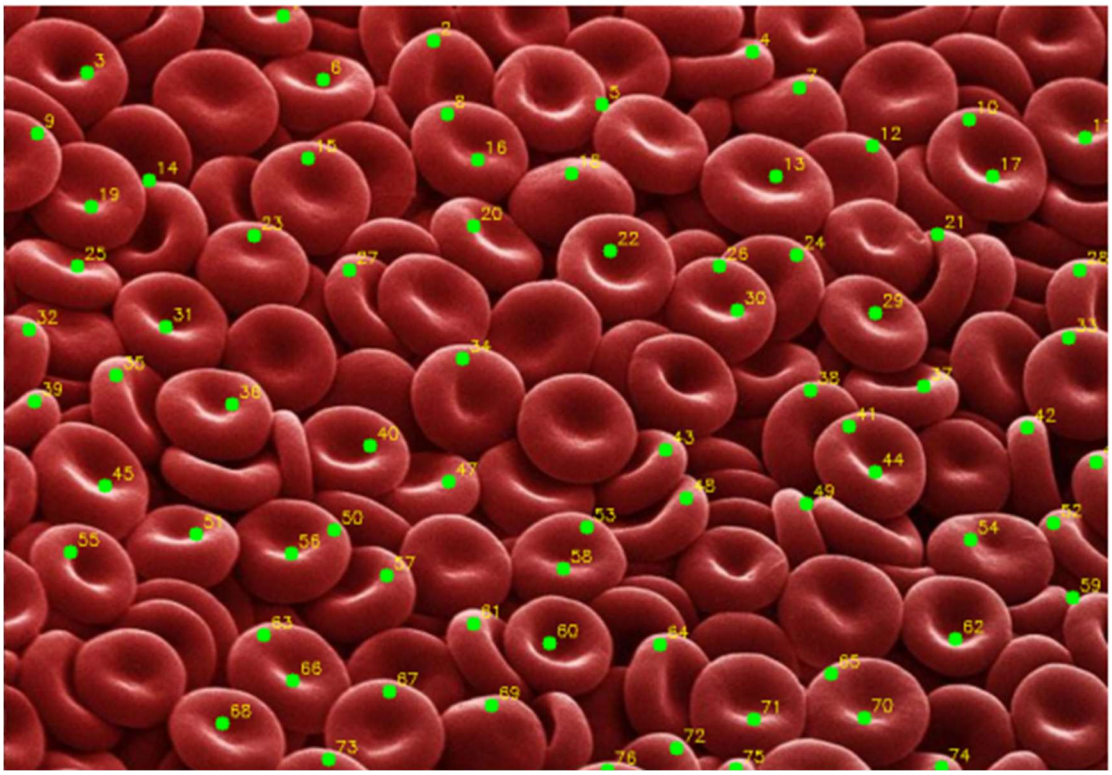
After opening and closing:



After eroding:



Final count:



So it counted most of them.

Closing Thoughts

Because the cells are often darker in the center than the areas they are separated from it, The thresholding is incredibly tricky. The only way I could find to count them was looking at the shiny spots, but that inevitably either double counts something after eroding or blends cells together after dilating.